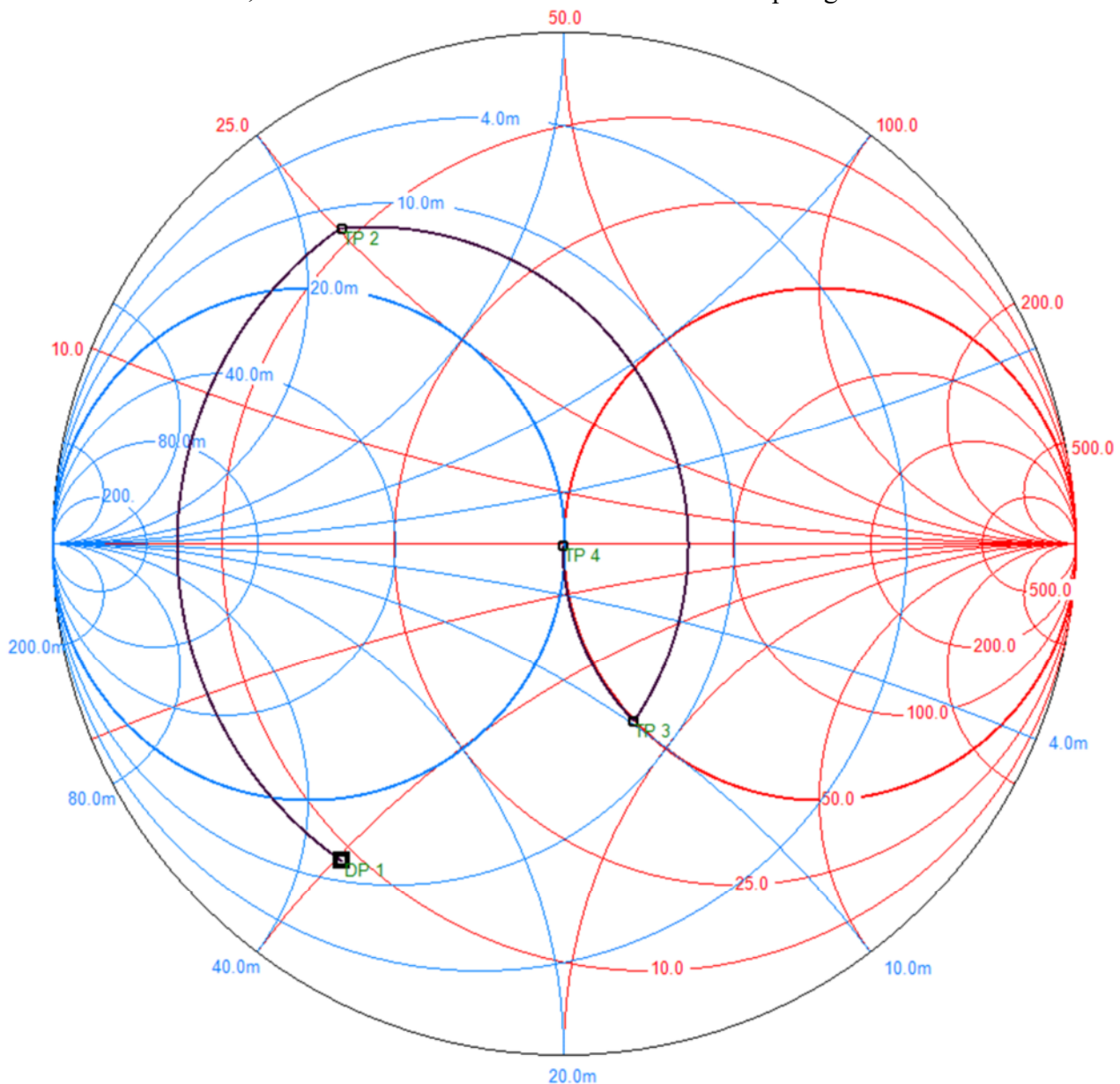


Aufgabe 1: (18 Punkte)

$$1. \quad \lambda = \frac{\lambda_0}{\sqrt{\epsilon_{r,eff}}} = \frac{c_0}{f \cdot \sqrt{\epsilon_{r,eff}}} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{1,8 \cdot 10^9 \frac{1}{\text{s}} \cdot \sqrt{3,3}} = 91,75 \text{ mm}$$

$$\frac{l_1}{\lambda} = \frac{14,0 \text{ mm}}{91,75 \text{ mm}} = 0,1526$$

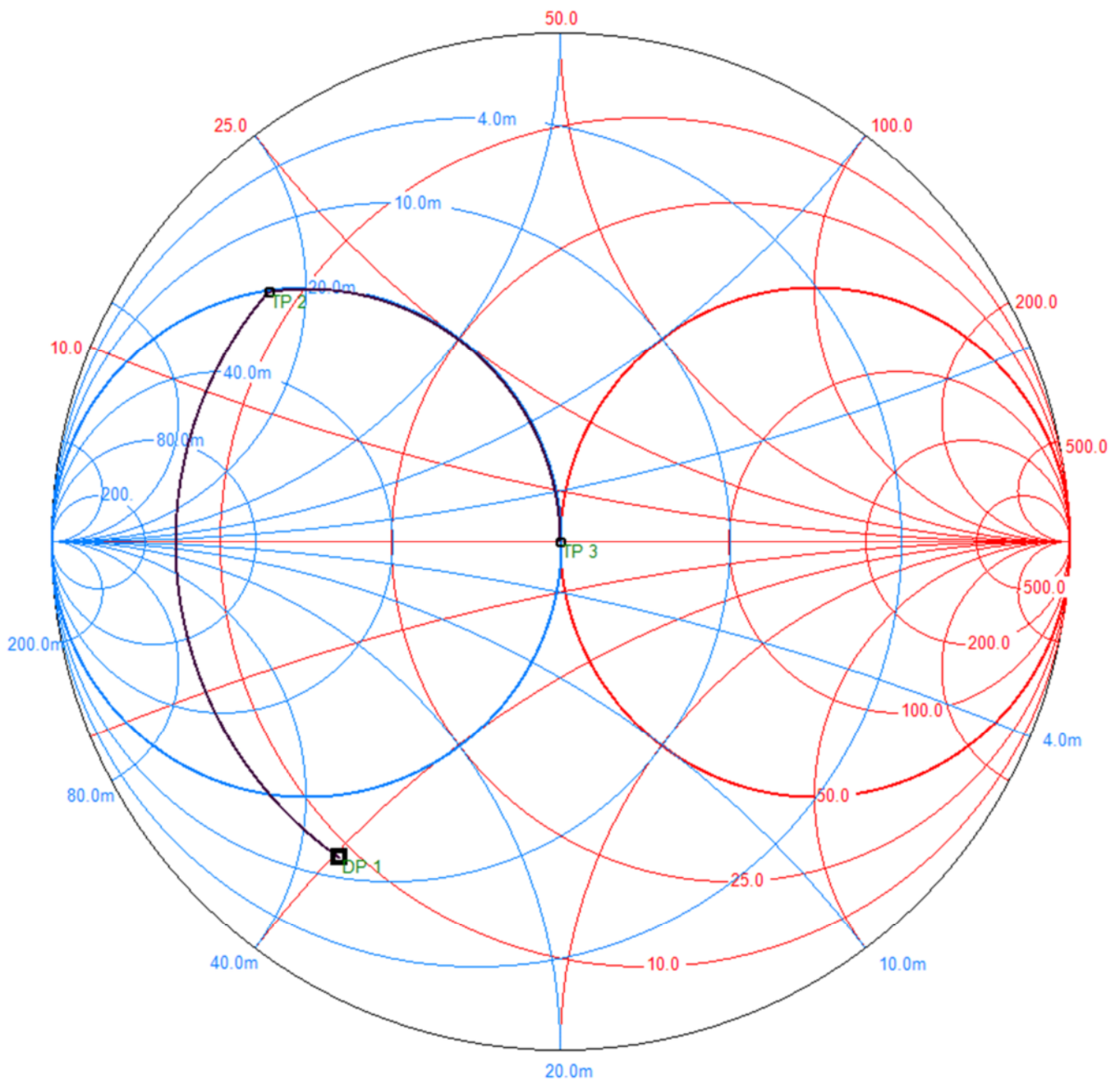
$\underline{s}_{11}$ einzeichnen in Smith-Chart, um l_1 transformieren, Parallel-C bis Schnittpunkt Einheitskreis, dann Transformation um Serien-L in den Ursprung



$$\text{ablesen: } b_c = 0,487 - (-1,79) = 2,277 \rightarrow B_c = \frac{b_c}{Z_L} = 45,54 \text{ mS} \quad C = \frac{B_c}{\omega} = \frac{45,54 \cdot 10^{-3} \frac{\text{A}}{\text{V}}}{2\pi \cdot 1,8 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{4,03 \text{ pF}}}$$

$$x_L = 0 - (-0,80) = 0,80 \rightarrow X_L = x_L \cdot Z_L = 40 \Omega \quad L = \frac{X_L}{\omega} = \frac{40 \frac{\text{V}}{\text{A}}}{2\pi \cdot 1,8 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{3,54 \text{ nH}}}$$

2. Transformation $\underline{S}_{11}$ über Leitung l_1' bis Leitwert-Einheitskreis, dann Transformation um C' in den Ursprung:



ablesen:

$$\frac{l_1'}{\lambda} = 0,05657 - 0,4236 (+0,5) = 0,133$$

$$b_c = 0 - (-2,29) = 2,29 \rightarrow B_c = \frac{b_c}{Z_L} = 45,8 \text{ mS}$$

$$\rightarrow \underline{l_1} = 0,33 \cdot 91,75 \text{ mm} = \underline{\underline{12,2 \text{ mm}}}$$

$$\underline{\underline{C}} = \frac{B_c}{\omega} = \frac{45,8 \cdot 10^{-3} \frac{\text{A}}{\text{V}}}{2\pi \cdot 1,8 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{4,05 \text{ pF}}}$$

Aufgabe 2:

(15 Punkte)

1. $\frac{w}{h} = 0,25$ ablesen aus Folie 3-5: $\varepsilon_{r,eff} = 3,2$

$$\lambda = \frac{\lambda_0}{\sqrt{\varepsilon_{r,eff}}} = \frac{c_0}{f \cdot \sqrt{\varepsilon_{r,eff}}} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{2,0 \cdot 10^9 \frac{1}{\text{s}} \cdot \sqrt{3,2}} = 83,85 \text{mm}$$

2. abgelesen aus Folie 3-6 mit $\frac{w}{h} = 0,25$: $Z_L = 110\Omega$

3. $\frac{l}{\lambda} = \frac{105 \text{mm}}{84 \text{mm}} = 1,25$ eine Wellenlänge abziehen: $\rightarrow \frac{\lambda}{4}$

damit aus Folie 7-14 mit $\frac{s}{h} = \frac{0,20 \text{mm}}{1,0 \text{mm}} = 0,2$: $C = C_{\min} = 7 \text{dB}$

alternativ:

mit Folie 7-12 und $\frac{s}{h} = 0,2$: $k_L = 0,44$ 1P

$$\begin{aligned} |\underline{s}_{41}|^2 &= \frac{k_L^2 \cdot \sin^2 \beta l}{(1 - k_L^2) \cdot \cos^2 \beta l + \sin^2 \beta l} = \frac{k_L^2 \cdot \sin^2 \beta l}{-k_L^2 \cdot \cos^2 \beta l + \cos^2 \beta l + \sin^2 \beta l} = \\ &= \frac{k_L^2 \cdot \sin^2 \beta l}{1 - k_L^2 \cdot \cos^2 \beta l} = \frac{(0,44)^2 \cdot \sin^2 \left(\frac{2\pi}{84 \text{mm}} \cdot 105 \text{mm} \right)}{1 - (0,44)^2 \cdot \cos^2 \left(\frac{2\pi}{84 \text{mm}} \cdot 105 \text{mm} \right)} = 0,1936 \end{aligned}$$

$$\underline{\underline{C}} = -10 \text{dB} \cdot \lg(|\underline{s}_{41}|^2) = -10 \text{dB} \cdot \lg(0,1936) = \underline{\underline{7,1 \text{dB}}}$$

4. maximale Koppeldämpfung $\cong$ keine Kopplung

$$\rightarrow l = \frac{\lambda'}{2} \quad \lambda' = 2 \cdot l$$

$$\underline{\underline{f'}} = \frac{c_0}{\lambda' \cdot \sqrt{\varepsilon_{r,eff}}} = \frac{c_0}{2 \cdot l \cdot \sqrt{\varepsilon_{r,eff}}} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{2 \cdot 105 \cdot 10^{-3} \text{m} \cdot \sqrt{3,2}} = \underline{\underline{799 \text{MHz}}}$$

Aufgabe 3:

(20 Punkte)

$$1. \quad \underline{s}_{11} = \frac{\underline{Z}_e - \underline{Z}_L}{\underline{Z}_e + \underline{Z}_L} = \frac{25\Omega \cdot e^{j62^\circ} - 50\Omega}{25\Omega \cdot e^{j62^\circ} + 50\Omega} = 0,6737 \cdot e^{130,3^\circ}$$

$$\underline{s}_{22} = \frac{\underline{Z}_a - \underline{Z}_L}{\underline{Z}_a + \underline{Z}_L} = \frac{36\Omega \cdot e^{-j23^\circ} - 50\Omega}{36\Omega \cdot e^{-j23^\circ} + 50\Omega} = 0,2604 \cdot e^{-130,6^\circ}$$

$$2. \quad |\underline{s}_{21}| = 10^{\frac{7,0\text{dB}}{20\text{dB}}} = 2,239$$

$$\underline{v}_{Tr,max} = 10\text{dB} \cdot \lg \left(|\underline{s}_{21}|^2 \cdot \frac{1}{1 - |\underline{s}_{11}|^2} \cdot \frac{1}{1 - |\underline{s}_{22}|^2} \right) = 10\text{dB} \cdot \lg \left(2,239^2 \cdot \frac{1}{1 - 0,67^2} \cdot \frac{1}{1 - 0,26^2} \right) = \underline{\underline{9,9\text{dB}}}$$

$$3. \quad |\underline{s}_{21}| = 2,239 \text{ (siehe Teilaufgabe 3)} \quad |\underline{s}_{12}| = 10^{\frac{-23\text{dB}}{20\text{dB}}} = 0,0708$$

ungünstigste Phasenlage für K : $|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|$ wird minimal,

d.h. $|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}| = |\underline{s}_{11}\underline{s}_{22}| - |\underline{s}_{12}\underline{s}_{21}| = 0,6737 \cdot 0,2604 - 2,239 \cdot 0,0708 = 0,0169 \quad [0,0157]$

$$K_{min} = \frac{1 - |\underline{s}_{11}|^2 - |\underline{s}_{22}|^2 + |\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|_{min}^2}{2|\underline{s}_{12}\underline{s}_{21}|} =$$

$$= \frac{1 - 0,6737^2 - 0,2604^2 + 0,0169^2}{2 \cdot 0,0708 \cdot 2,239} = 1,510 > 1 \quad \text{"stabil"} \quad [1,526]$$

ungünstigste Phasenlage für B : $|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|$ wird maximal,

d.h. $|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}| = |\underline{s}_{11}\underline{s}_{22}| + |\underline{s}_{12}\underline{s}_{21}| = 0,6737 \cdot 0,2604 + 2,239 \cdot 0,0708 = 0,3339 \quad [0,3327]$

$$B_{min} = 1 + |\underline{s}_{11}|^2 - |\underline{s}_{22}|^2 - |\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|_{max}^2 =$$

$$= 1 + 0,6737^2 - 0,2604^2 - 0,3339^2 = 1,275 > 0 \quad \text{"stabil"} \quad [1,271]$$

→ Der Transistor ist auf jeden Fall stabil!

Aufgabe 4:

(18 Punkte)

$$1. \quad p_A + g_{VV1} - a_{L1} + g_{VV2,min} - a_{L2} = p_{E,min}$$

$$\underline{\underline{g_{VV2,min} = p_{E,min} - p_A - g_{VV1} + a_{L1} + a_{L2} = -65\text{dBm} - (-70\text{dBm}) - 15\text{dB} + 20\text{dB} + 3\text{dB} = 13\text{dB}}}$$

2.

$$P_{R,Ant} = k \cdot T \cdot B = 1,38 \cdot 10^{-23} \frac{\text{Ws}}{\text{K}} \cdot 290\text{K} \cdot 50 \cdot 10^6 \frac{1}{\text{s}} = 2,0 \cdot 10^{-13} \text{W} \hat{=} -97\text{dBm}$$

$$\left(\frac{S}{N} \right)_{\text{ein}} = \left(\frac{P_A}{P_{R,Ant}} \right) \hat{=} -70\text{dBm} - (-97\text{dBm}) = 27\text{dB}$$

$$F_{\text{ges,max}} = \frac{\left(\frac{S}{N} \right)_{\text{ein}}}{\left(\frac{S}{N} \right)_{\text{aus}}} \hat{=} 27\text{dB} - 18\text{dB} = 9,0\text{dB} = f_{\text{ges}}$$

3.

$$G_{VV1} = 10^{\frac{g_{VV1}}{10\text{dB}}} = 10^{\frac{15\text{dB}}{10\text{dB}}} = 31,6 \quad F_{VV1} = 10^{\frac{f_{VV1}}{10\text{dB}}} = 10^{\frac{1,5\text{dB}}{10\text{dB}}} = 1,41$$

$$G_{L1} = 10^{-\frac{a_{L1}}{10\text{dB}}} = 10^{-\frac{20\text{dB}}{10\text{dB}}} = 0,01 \quad F_{L1} = 10^{\frac{a_{L1}}{10\text{dB}}} = 10^{\frac{20\text{dB}}{10\text{dB}}} = 100$$

$$G_{VV2} = 10^{\frac{g_{VV2}}{10\text{dB}}} = 10^{\frac{15\text{dB}}{10\text{dB}}} = 31,6 \quad F_{VV2} = ?$$

$$F_{L2} = 10^{\frac{a_{L1}}{10\text{dB}}} = 10^{\frac{3,0\text{dB}}{10\text{dB}}} = 2,0$$

$$F_{\text{ges}} = 10^{\frac{9,0\text{dB}}{10\text{dB}}} = 7,94$$

$$F_{\text{ges}} = 1 + (F_{VV1} - 1) + \frac{F_{L1} - 1}{G_{VV2}} + \frac{F_{VV2} - 1}{G_{VV1} \cdot G_{L1}} + \frac{F_{L2} - 1}{G_{VV1} \cdot G_{L1} \cdot G_{VV2}}$$

$$\frac{F_{VV2} - 1}{G_{VV1} \cdot G_{L1}} = F_{\text{ges}} - F_{VV1} - \frac{F_{L1} - 1}{G_{VV2}} - \frac{F_{L2} - 1}{G_{VV1} \cdot G_{L1} \cdot G_{VV2}}$$

$$F_{VV2} = 1 + G_{VV1} \cdot G_{L1} \cdot \left(F_{\text{ges}} - F_{VV1} - \frac{F_{L1} - 1}{G_{VV2}} - \frac{F_{L2} - 1}{G_{VV1} \cdot G_{L1} \cdot G_{VV2}} \right) =$$

$$= 1 + 31,6 \cdot 0,01 \cdot \left(7,94 - 1,41 - \frac{100 - 1}{31,6} - \frac{2,0 - 1}{31,6 \cdot 0,01 \cdot 31,6} \right) = 2,042$$

$$\underline{\underline{f_{VV2} = 10\text{dB} \cdot \lg F_{VV2} = 3,1\text{dB}}}$$

Aufgabe 5:

(21 Punkte)

1.

$$P_{Q,max} = \frac{1}{2} |\underline{b}_Q|^2 \cdot \frac{1}{1 - |\underline{r}_Q|^2} = P_{Q,50\Omega} \cdot \frac{1}{1 - |\underline{r}_Q|^2}$$

$$\underline{\underline{P_{Q,50\Omega}}} = P_{Q,max} \cdot (1 - |\underline{r}_Q|^2) = 4,0 \text{ W} \cdot (1 - 0,8^2) = \underline{\underline{1,44 \text{ W}}}$$

2.

$$\underline{r}_2 = \underline{s}_{22} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_Q}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = j0,30 + \frac{0,7 \cdot 0,7 \cdot j0,8}{1 - j0,30 \cdot j0,8} = j0,6161 = 0,6161 \cdot e^{j90^\circ}$$

$$\underline{\underline{r_L}} = \underline{r}_2^* = \underline{\underline{0,6161 \cdot e^{-j90^\circ}}}$$

3.

$$\underline{b}_{Q2} = \underline{b}_Q \cdot \frac{\underline{s}_{21}}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = \underline{b}_Q \cdot \frac{0,70}{1 - j0,30 \cdot j0,8} = \underline{b}_Q \cdot 0,5645$$

$$\underline{\underline{P_L}} = \frac{1}{2} |\underline{b}_{Q2}|^2 \cdot \frac{1 - |\underline{r}_L|^2}{|1 - \underline{r}_2 \cdot \underline{r}_L|^2} = \frac{1}{2} |\underline{b}_Q|^2 \cdot 0,5645^2 \cdot \frac{1 - 0,55^2}{|1 - j0,60 \cdot (-j0,55)|^2} = 1,5 \text{ W} \cdot 0,4536 = \underline{\underline{0,743 \text{ W}}}$$

4.

$$P_{L,abs,0} = \frac{1}{2} |\underline{b}_Q|^2 \cdot \frac{1 - |\underline{r}_L|^2}{|1 - \underline{r}_Q \cdot \underline{r}_L|^2} = 1,5 \text{ W} \cdot \frac{1 - 0,55^2}{|1 - j0,8 \cdot (-j0,55)|^2} = 3,34 \text{ W}$$

$$\underline{\underline{a_E}} = 10 \text{ dB} \cdot \lg\left(\frac{P_{L,abs,0}}{P_L}\right) = 10 \text{ dB} \cdot \lg\left(\frac{3,34 \text{ W}}{0,743 \text{ W}}\right) = \underline{\underline{6,5 \text{ dB}}} \quad [13,9 \text{ dB}]$$

Reflexionsfaktor der Ersatzlast:

$$\underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L}{1 - \underline{s}_{22} \cdot \underline{r}_L} = j0,30 + \frac{0,7 \cdot 0,7 \cdot (-j0,55)}{1 - j0,30 \cdot (-j0,55)} = -j0,02275$$

$$P_{Q,ab} = \frac{1}{2} |\underline{b}_Q|^2 \cdot \frac{1 - |\underline{r}_1|^2}{|1 - \underline{r}_Q \cdot \underline{r}_1|^2} = 1,5 \text{ W} \cdot \frac{1 - 0,02275^2}{|1 - j0,8 \cdot (-j0,02275)|^2} = 1,53 \text{ W}$$

$$\underline{\underline{a_B}} = 10 \text{ dB} \cdot \lg\left(\frac{P_{Q,ab}}{P_{L,max}}\right) = 10 \text{ dB} \cdot \lg\left(\frac{1,53 \text{ W}}{0,743 \text{ W}}\right) = \underline{\underline{3,1 \text{ dB}}} \quad [10,5 \text{ dB}]$$